

1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI} :

$$\|A\mathbf{r}_1(E_1) - B\mathbf{r}_2(E_2)\| = r_{\text{SOI}} \quad (1)$$

where A and B are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i : semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i \sqrt{1 - e_i^2}$)
- E_i : eccentric anomaly of orbit i
- $A \in \mathbb{R}^{3 \times 2}, B \in \mathbb{R}^{3 \times 2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI} : radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\mathbf{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \quad (2)$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(A\mathbf{r}_1 - B\mathbf{r}_2)^\top (A\mathbf{r}_1 - B\mathbf{r}_2) = r_{\text{SOI}}^2 \quad (3)$$

Expand the left-hand side:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - 2\mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (4)$$

1.4.1. Orthonormality of A and B

Since A and B are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$A^\top A = I_{2 \times 2}, \quad B^\top B = I_{2 \times 2} \quad (5)$$

Therefore the self-terms simplify:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 = \mathbf{r}_1^\top \mathbf{r}_1 = |\mathbf{r}_1|^2 =: r_1^2 \quad (6)$$

$$\mathbf{r}_2^\top B^\top B \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_2 = |\mathbf{r}_2|^2 =: r_2^2 \quad (7)$$

1.4.2. The coupling matrix C

Define the 2×2 matrix:

$$C := 2A^\top B \quad (8)$$

This encodes the entire mutual orientation of the two orbital planes. Equation 4 becomes:

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (9)$$

Note: C is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

1.5. Newton's Method

We seek to minimise $f(E_1, E_2) := r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 - r_{\text{SOI}}^2$ using Newton's method. Each iteration solves the 2×2 linear system $H\delta = -\nabla f$ for the step $\delta = (\delta E_1, \delta E_2)^\top$, then updates $E_i \leftarrow E_i + \delta E_i$.

1.5.1. Deriving the gradient and Hessian

The objective is $f = r_1^2 + r_2^2 - \mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 - r_{\text{SOI}}^2$, with r_{SOI}^2 constant. We differentiate each part with respect to E_i .

1.5.1.1. Self-terms r_i^2

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$\frac{\partial r_i^2}{\partial E_i} = 2a_i^2 e_i \sin E_i (1 - e_i \cos E_i) \quad (10)$$

$$\frac{\partial^2 r_i^2}{\partial E_i^2} = 2a_i^2 e_i (\cos E_i - e_i + 2e_i \sin^2 E_i) \quad (11)$$

and $\partial r_i^2 / \partial E_j = 0$ for $i \neq j$.

1.5.1.2. Cross-term $\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2$

The position vector Equation 2 can be factored as $\mathbf{r}_i = \text{diag}(a_i, b_i) \mathbf{p}_i$, where $\text{diag}(a_i, b_i) := \begin{pmatrix} a_i & 0 \\ 0 & b_i \end{pmatrix}$ is a diagonal matrix and $\mathbf{p}_i$ is a dimensionless position-direction vector:

$$\mathbf{p}_i = \begin{pmatrix} \cos E_i - e_i \\ \sin E_i \end{pmatrix} \quad (12)$$

Substituting $\mathbf{r}_i = \text{diag}(a_i, b_i) \mathbf{p}_i$ into the cross-term and using $(\mathbf{X} \mathbf{p})^\top = \mathbf{p}^\top \mathbf{X}^\top$:

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = (\text{diag}(a_1, b_1) \mathbf{p}_1)^\top \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1)^\top \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 \quad (13)$$

Since diagonal matrices are symmetric ($\text{diag}(x, y)^\top = \text{diag}(x, y)$):

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 \quad (14)$$

We have a chain of five matrices. Applying associativity ($\mathbf{A}(\mathbf{B}\mathbf{C}) = (\mathbf{A}\mathbf{B})\mathbf{C}$) twice – first to $(\mathbf{p}_1^\top \cdot \text{diag}(a_1, b_1)) \cdot \mathbf{C} = \mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot \mathbf{C})$, then to $(\mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot \mathbf{C})) \cdot \text{diag}(a_2, b_2) = \mathbf{p}_1^\top \cdot ((\text{diag}(a_1, b_1) \cdot \mathbf{C}) \cdot \text{diag}(a_2, b_2))$ – lets us group the three constant middle factors into a single matrix. We define the **scaled coupling matrix**:

$$\mathbf{M} := \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) = \begin{pmatrix} C_{11}a_1a_2 & C_{12}a_1b_2 \\ C_{21}a_2b_1 & C_{22}b_1b_2 \end{pmatrix} \quad (15)$$

giving $\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$.

Since $\mathbf{M}$ is constant, differentiation acts only on $\mathbf{p}_1$ and $\mathbf{p}_2$. Define the derivative vectors:

$$\hat{\mathbf{w}}_i := \frac{d\mathbf{p}_i}{dE_i} = \begin{pmatrix} -\sin E_i \\ \cos E_i \end{pmatrix}, \quad \hat{\mathbf{u}}_i := \begin{pmatrix} \cos E_i \\ \sin E_i \end{pmatrix} \quad (16)$$

To differentiate $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$ with respect to E_1 , note that only $\mathbf{p}_1$ depends on E_1 . By associativity, $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = \mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2)$, which is a dot product of $\mathbf{p}_1$ with the constant vector $\mathbf{M} \mathbf{p}_2$. The derivative of a dot product with one constant factor is:

$$\frac{\partial(\mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2))}{\partial E_1} = \hat{\mathbf{w}}_1^\top (\mathbf{M} \mathbf{p}_2) = \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (17)$$

Similarly, for E_2 only $\mathbf{p}_2$ depends on E_2 . By associativity, $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = (\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2$, a dot product of the constant row vector $\mathbf{p}_1^\top \mathbf{M}$ with $\mathbf{p}_2$:

$$\frac{\partial((\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2)}{\partial E_2} = (\mathbf{p}_1^\top \mathbf{M}) \hat{\mathbf{w}}_2 = \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (18)$$

The same reasoning applies to higher derivatives, replacing $\mathbf{p}_i$ by successive derivatives $\hat{\mathbf{w}}_i, -\hat{\mathbf{u}}_i$:

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_1^2} = -\hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2, \quad \frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_2^2} = -\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2 \quad (19)$$

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_1 \partial E_2} = \hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (20)$$

1.5.1.3. Combined expressions

Combining $f = r_1^2 + r_2^2 - \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 - r_{\text{SOI}}^2$ using Equation 10 with Equation 17, and Equation 11 with Equation 19 and Equation 20:

1.5.2. Gradient

$$\frac{\partial f}{\partial E_1} = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (21)$$

$$\frac{\partial f}{\partial E_2} = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (22)$$

1.5.3. Hessian

The Hessian $\mathbf{H}$ is symmetric ($H_{12} = H_{21}$):

$$H_{11} = 2a_1^2 e_1 (\cos E_1 - e_1 + 2e_1 \sin^2 E_1) + \hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (23)$$

$$H_{22} = 2a_2^2 e_2 (\cos E_2 - e_2 + 2e_2 \sin^2 E_2) + \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2 \quad (24)$$

$$H_{12} = -\hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (25)$$

1.5.4. Solving the Newton step

The 2×2 system is solved directly via Cramer's rule:

$$\Delta = H_{11} H_{22} - H_{12}^2 \quad (26)$$

$$\delta E_1 = \frac{\frac{\partial f}{\partial E_1} H_{22} - \frac{\partial f}{\partial E_2} H_{12}}{-\Delta}, \quad \delta E_2 = \frac{\frac{\partial f}{\partial E_2} H_{11} - \frac{\partial f}{\partial E_1} H_{12}}{-\Delta} \quad (27)$$

All coupling terms are bilinear forms $\mathbf{x}^\top \mathbf{M} \mathbf{y}$ with 2D vectors, each requiring 4 multiplies and 3 adds. The five needed dot products ($\hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2$, $\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2$, $\hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2$, $\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2$, $\hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2$) can share the intermediate products $\mathbf{M} \mathbf{p}_2$, $\mathbf{M} \hat{\mathbf{w}}_2$, $\mathbf{M} \hat{\mathbf{u}}_2$ (each a 2D matrix-vector multiply).

1.6. Coarse Encounter Interval

Before running Newton's method, we restrict the search to E_1 values where the spacecraft's heliocentric distance overlaps with the planet's distance range $\pm r_{\text{SOI}}$.

1.6.1. Radial overlap condition

The heliocentric distance of the spacecraft is $r_1 = a_1(1 - e_1 \cos E_1)$. An encounter requires:

$$a_2(1 - e_2) - r_{\text{SOI}} \leq r_1 \leq a_2(1 + e_2) + r_{\text{SOI}} \quad (28)$$

Substituting $r_1 = a_1(1 - e_1 \cos E_1)$ and solving for $\cos E_1$:

$$\frac{a_1 - a_2(1 + e_2) - r_{\text{SOI}}}{a_1 e_1} \leq \cos E_1 \leq \frac{a_1 - a_2(1 - e_2) + r_{\text{SOI}}}{a_1 e_1} \quad (29)$$

Clamping both sides to $[-1, 1]$ and applying arccos (which reverses the inequality) gives two symmetric intervals $[E_{\text{lo}}, E_{\text{hi}}]$ and $[-E_{\text{lo}}, -E_{\text{hi}}]$ in E_1 . If no valid interval exists (i.e. the clamped range is empty), no encounter is possible at any E_1 .

1.6.2. Initial guess via parabolic interpolation

For each of the two symmetric E_1 intervals, we evaluate f at three points: the endpoints and midpoint. Here E_2 is estimated by projecting the spacecraft's 3D position onto the planet's orbital plane and inverting the ellipse parametrisation:

$$E_2 = \text{atan2}(q_y/b_2, q_x/a_2 + e_2) \quad (30)$$

where $\mathbf{q} = \mathbf{B}^\top \mathbf{A} \mathbf{r}_1$ is the projection. This matches the eccentric anomaly that $\mathbf{q}$ would have if it lay on the planet's ellipse.

Given three samples (x_0, f_0) , (x_1, f_1) , (x_2, f_2) , the minimiser of the interpolating quadratic is:

$$E_1^* = x_1 - \frac{1}{2} \frac{(x_1 - x_0)^2(f_1 - f_2) - (x_1 - x_2)^2(f_1 - f_0)}{(x_1 - x_0)(f_1 - f_2) - (x_1 - x_2)(f_1 - f_0)} \quad (31)$$

Newton's method is then run from the best initial guess across both branches. If the first branch fails to converge to $f \leq r_{\text{SOI}}^2$, the second branch is tried.

1.7. Encounter Intervals in E_1 and E_2

The constraint $f(E_1, E_2) = 0$ defines an implicit curve (or set of curves) in the (E_1, E_2) plane. The **encounter interval** $[E_{1,\min}, E_{1,\max}]$ is the projection of this curve onto the E_1 axis, and $[E_{2,\min}, E_{2,\max}]$ is the projection onto the E_2 axis. For every E_1 in the first interval there exists at least one E_2 in the second interval such that the distance equals r_{SOI} .

1.7.1. Extrema via implicit differentiation

Along the constraint curve $f(E_1, E_2) = 0$, total differentiation gives:

$$\frac{\partial f}{\partial E_1} dE_1 + \frac{\partial f}{\partial E_2} dE_2 = 0 \quad (32)$$

so the implicit derivatives are:

$$\frac{dE_1}{dE_2} = -\frac{\partial f / \partial E_2}{\partial f / \partial E_1}, \quad \frac{dE_2}{dE_1} = -\frac{\partial f / \partial E_1}{\partial f / \partial E_2} \quad (33)$$

The extrema of E_1 along the constraint curve occur where the tangent is vertical, i.e. $dE_1 / dE_2 = 0$, which requires:

$$\frac{\partial f}{\partial E_2} = 0 \quad (34)$$

Similarly, the extrema of E_2 occur where $dE_2 / dE_1 = 0$, requiring:

$$\frac{\partial f}{\partial E_1} = 0 \quad (35)$$

1.7.2. System for E_1 bounds

To find $E_{1,\min}$ and $E_{1,\max}$, solve the 2×2 system:

$$\begin{cases} f(E_1, E_2) = 0 \\ \frac{\partial f}{\partial E_2} = 0 \end{cases} \quad (36)$$

Using Equation 22, the second equation is:

$$2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 = 0 \quad (37)$$

Newton's method for this system uses the Jacobian:

$$\mathbf{J}_1 = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial^2 f}{\partial E_1 \partial E_2} & \frac{\partial^2 f}{\partial E_2^2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ H_{12} & H_{22} \end{pmatrix} \quad (38)$$

where $g_1 := \partial f / \partial E_1$ (Equation 21), $g_2 := \partial f / \partial E_2$ (Equation 22), and H_{12}, H_{22} are the Hessian entries (Equation 25, Equation 24). The Newton step $\boldsymbol{\delta} = (\delta E_1, \delta E_2)^\top$ solves $\mathbf{J}_1 \boldsymbol{\delta} = -(\mathbf{f}, g_2)^\top$ via Cramer's rule:

$$\Delta_1 = g_1 H_{22} - g_2 H_{12} \quad (39)$$

$$\delta E_1 = \frac{f H_{22} - g_2^2}{\Delta_1}, \quad \delta E_2 = \frac{f H_{12} - g_1 g_2}{\Delta_1} \quad (40)$$

Note that at convergence ($g_2 = 0$), the determinant simplifies to $\Delta_1 = g_1 H_{22}$.

1.7.3. System for E_2 bounds

To find $E_{2,\min}$ and $E_{2,\max}$, solve:

$$\begin{cases} f(E_1, E_2) = 0 \\ \frac{\partial f}{\partial E_1} = 0 \end{cases} \quad (41)$$

Using Equation 21, the second equation is:

$$2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 = 0 \quad (42)$$

The Jacobian is:

$$\mathbf{J}_2 = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial^2 f}{\partial E_1^2} & \frac{\partial^2 f}{\partial E_1 \partial E_2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ H_{11} & H_{12} \end{pmatrix} \quad (43)$$

The Newton step solves $\mathbf{J}_2 \boldsymbol{\delta} = -(\mathbf{f}, g_1)^\top$:

$$\Delta_2 = g_1 H_{12} - g_2 H_{11} \quad (44)$$

$$\delta E_1 = \frac{f H_{12} - g_1 g_2}{\Delta_2}, \quad \delta E_2 = \frac{g_1^2 - f H_{11}}{\Delta_2} \quad (45)$$

At convergence ($g_1 = 0$), the determinant simplifies to $\Delta_2 = -g_2 H_{11}$.

1.7.4. Geometric interpretation

The constraint curve $f = 0$ forms a closed loop (or multiple loops) in the (E_1, E_2) torus. The critical points Equation 34 are where the curve has a vertical tangent — E_1 reaches a turning point as E_2 varies. The two solutions give $E_{1, \min}$ and $E_{1, \max}$. Likewise, Equation 35 gives horizontal tangents where E_2 is extremal, yielding $E_{2, \min}$ and $E_{2, \max}$.

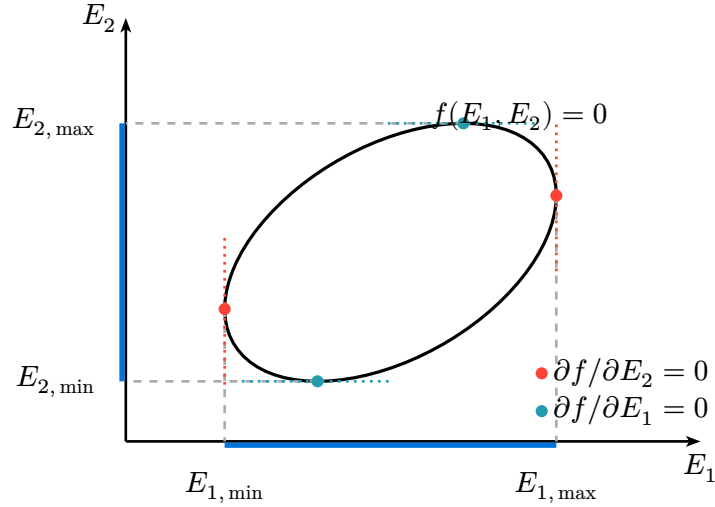


Figure 1: The constraint curve $f = 0$ in the (E_1, E_2) plane. Red dots mark vertical tangents ($\partial f / \partial E_2 = 0$), whose E_1 coordinates give $E_{1, \min}$ and $E_{1, \max}$. Teal dots mark horizontal tangents ($\partial f / \partial E_1 = 0$), whose E_2 coordinates give $E_{2, \min}$ and $E_{2, \max}$. Blue bars on the axes show the projected encounter intervals.

2. Time Constraint via Kepler's Equation

The preceding sections treated E_1 and E_2 as independent variables. In reality, both bodies obey Kepler's equation, which couples each eccentric anomaly to a common time t .

2.1. Kepler's equation

For each orbit i with mean motion $n_i = 2\pi/T_i$ (where T_i is the orbital period) and mean anomaly at epoch $M_{i,0}$:

$$M_i(t) = M_{i,0} + n_i t = E_i - e_i \sin E_i \quad (46)$$

This implicitly defines $E_i(t)$. At a given time t , both eccentric anomalies are determined:

$$E_1 - e_1 \sin E_1 = M_{1,0} + n_1 t, \quad E_2 - e_2 \sin E_2 = M_{2,0} + n_2 t \quad (47)$$

2.2. Eliminating time

The position on an ellipse is 2π -periodic in E_i , so the spacecraft returns to the same point after each full orbit. The set of times at which orbit i passes through eccentric anomaly $E_i \in [-\pi, \pi]$ is:

$$t = \frac{E_i - e_i \sin E_i - M_{i,0} + 2\pi k_i}{n_i}, \quad k_i \in \mathbb{Z} \quad (48)$$

For an encounter, both bodies must be at their respective positions **simultaneously**. Equating the two time expressions:

$$\frac{E_1 - e_1 \sin E_1 - M_{1,0} + 2\pi k_1}{n_1} = \frac{E_2 - e_2 \sin E_2 - M_{2,0} + 2\pi k_2}{n_2} \quad (49)$$

Rearranging gives a family of **time-coupling constraints**, one for each integer pair (k_1, k_2) :

$$h_{k_1, k_2}(E_1, E_2) := n_2(E_1 - e_1 \sin E_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) + 2\pi(n_2 k_1 - n_1 k_2) = 0 \quad (50)$$

Each choice of (k_1, k_2) corresponds to a different encounter opportunity (i.e. orbit 1 on its k_1 -th revolution meeting orbit 2 on its k_2 -th revolution). Since only the combination $n_2 k_1 - n_1 k_2$ appears, the distinct constraints are parametrised by a single offset:

$$h(E_1, E_2; \lambda) := n_2(E_1 - e_1 \sin E_1 - M_{1,0}) - n_1(E_2 - e_2 \sin E_2 - M_{2,0}) + 2\pi\lambda = 0 \quad (51)$$

where $\lambda = n_2 k_1 - n_1 k_2$ ranges over a discrete set. In the (E_1, E_2) plane (with $E_i \in [-\pi, \pi]$), each value of λ gives a monotone curve from bottom-left to top-right, and these curves are spaced apart by the synodic offset. The full encounter problem is: find (E_1, E_2) satisfying both the distance constraint $f = 0$ (Equation 9) and $h = 0$ (Equation 51) for some admissible λ .

2.3. Enumerating encounter opportunities

Since E_i and $\sin E_i$ are 2π -periodic, shifting k_2 by 1 shifts λ by $-n_1$, which shifts E_2 by exactly 2π — a full orbit. On the torus $E_i \in [-\pi, \pi]$, different k_2 values for the same k_1 therefore produce the **same** curve. The distinct encounter opportunities are parametrised by k_1 alone.

2.3.1. Finding the k_1 range from a time window

Given a time window $[t_{lo}, t_{hi}]$, the spacecraft passes through eccentric anomaly $E_1 \in [-\pi, \pi]$ on its k_1 -th orbit at time (Equation 48):

$$t = \frac{E_1 - e_1 \sin E_1 - M_{1,0} + 2\pi k_1}{n_1} \quad (52)$$

Since $E_1 - e_1 \sin E_1 \in [-\pi, \pi]$, the admissible k_1 values satisfy:

$$k_{1, \min} = \left\lfloor \frac{n_1 t_{lo} + M_{1,0} - \pi}{2\pi} \right\rfloor, \quad k_{1, \max} = \left\lceil \frac{n_1 t_{hi} + M_{1,0} + \pi}{2\pi} \right\rceil \quad (53)$$

2.3.2. Choosing k_2 for each k_1

For each k_1 , the approximate encounter time is $t \approx (-M_{1,0} + 2\pi k_1)/n_1$ (at $E_1 \approx 0$). The corresponding k_2 is simply:

$$k_2 = \text{round}\left(\frac{n_2 t + M_{2,0}}{2\pi}\right) \quad (54)$$

This gives $\lambda = n_2 k_1 - n_1 k_2$, and each k_1 in $[k_{1, \min}, k_{1, \max}]$ is a candidate encounter opportunity that can be refined with Newton's method.

2.4. Gradient of h

$$\frac{\partial h}{\partial E_1} = n_2(1 - e_1 \cos E_1), \quad \frac{\partial h}{\partial E_2} = -n_1(1 - e_2 \cos E_2) \quad (55)$$

Note that $1 - e_i \cos E_i = r_i/a_i > 0$ for elliptic orbits, so $\partial h/\partial E_1 > 0$ and $\partial h/\partial E_2 < 0$ everywhere. The time constraint is therefore a monotone relation between E_1 and E_2 :

$$\frac{dE_2}{dE_1} = \frac{n_2(1 - e_1 \cos E_1)}{n_1(1 - e_2 \cos E_2)} = \frac{n_2 r_1 / a_1}{n_1 r_2 / a_2} > 0 \quad (56)$$

2.5. Newton's method on the combined system

For a given λ , we solve the 2×2 system $(f, h) = \mathbf{0}$ using Newton's method. The Jacobian is:

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial E_1} & \frac{\partial f}{\partial E_2} \\ \frac{\partial h}{\partial E_1} & \frac{\partial h}{\partial E_2} \end{pmatrix} = \begin{pmatrix} g_1 & g_2 \\ n_2(1 - e_1 \cos E_1) & -n_1(1 - e_2 \cos E_2) \end{pmatrix} \quad (57)$$

The Newton step $\delta = (\delta E_1, \delta E_2)^\top$ solves $\mathbf{J}\delta = -(f, h)^\top$ via Cramer's rule:

$$\Delta = -g_1 n_1 (1 - e_2 \cos E_2) - g_2 n_2 (1 - e_1 \cos E_1) \quad (58)$$

$$\delta E_1 = \frac{f n_1 (1 - e_2 \cos E_2) + g_2 h}{\Delta} \quad (59)$$

$$\delta E_2 = \frac{-f n_2 (1 - e_1 \cos E_1) - g_1 h}{\Delta} \quad (60)$$

where g_1, g_2 are the distance-constraint gradient components (Equation 21, Equation 22). Since $\partial h / \partial E_1 > 0$ and $\partial h / \partial E_2 < 0$ always, the Jacobian is non-singular whenever (g_1, g_2) is not parallel to $(n_2 r_1 / a_1, -n_1 r_2 / a_2)$.

2.6. Initial guess

For a given λ , pick $E_1^{(0)}$ as the midpoint of $[E_{1,\min}, E_{1,\max}]$. Compute the corresponding time from Equation 48 (with $k_1 = 0$):

$$t = \frac{E_1^{(0)} - e_1 \sin E_1^{(0)} - M_{1,0}}{n_1} \quad (61)$$

Rearranging the time constraint $h(E_1^{(0)}, E_2^{(0)}; \lambda) = 0$ (Equation 51) for E_2 's mean anomaly:

$$M_2^{(0)} := E_2^{(0)} - e_2 \sin E_2^{(0)} = M_{2,0} + n_2 t + \frac{2\pi\lambda}{n_1} \quad (62)$$

Substituting the expression for t :

$$M_2^{(0)} = M_{2,0} + \frac{n_2}{n_1} \left(E_1^{(0)} - e_1 \sin E_1^{(0)} - M_{1,0} \right) + \frac{2\pi\lambda}{n_1} \quad (63)$$

Then solve Kepler's equation $E_2^{(0)} - e_2 \sin E_2^{(0)} = M_2^{(0)}$ (e.g. by Newton iteration) to obtain $E_2^{(0)}$. This ensures the initial point lies exactly on the time-constraint curve for the chosen λ .

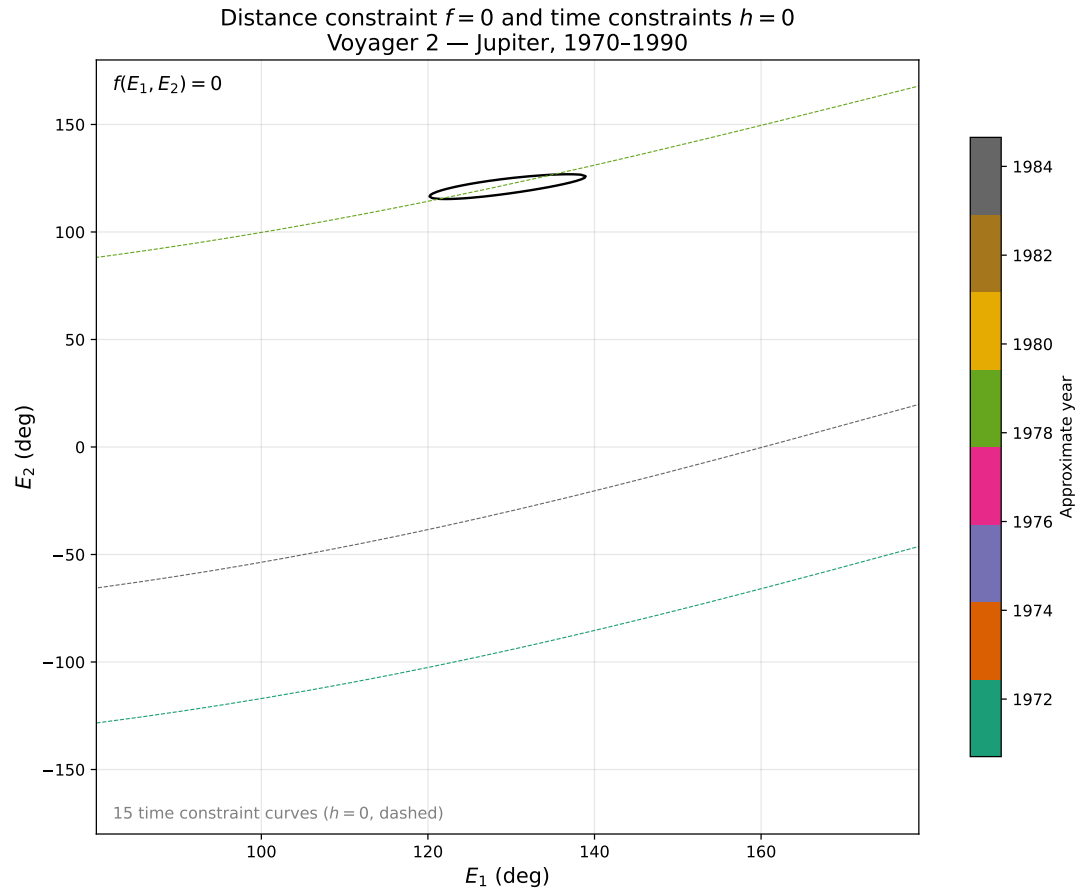


Figure 2: The distance constraint $f(E_1, E_2) = 0$ (solid black) and time constraints $h(E_1, E_2; \lambda) = 0$ (dashed, coloured by year) for the Voyager 2 — Jupiter system (1970–1990). Each dashed curve corresponds to a different encounter opportunity (synodic period ≈ 7 years). Intersections of the dashed curves with the solid curve are solutions to the full encounter problem.